UKE HAMBURG

German Conference on Bioinformatics

WS8: Spatial domain identification: computational methods for discovering tissue architecture

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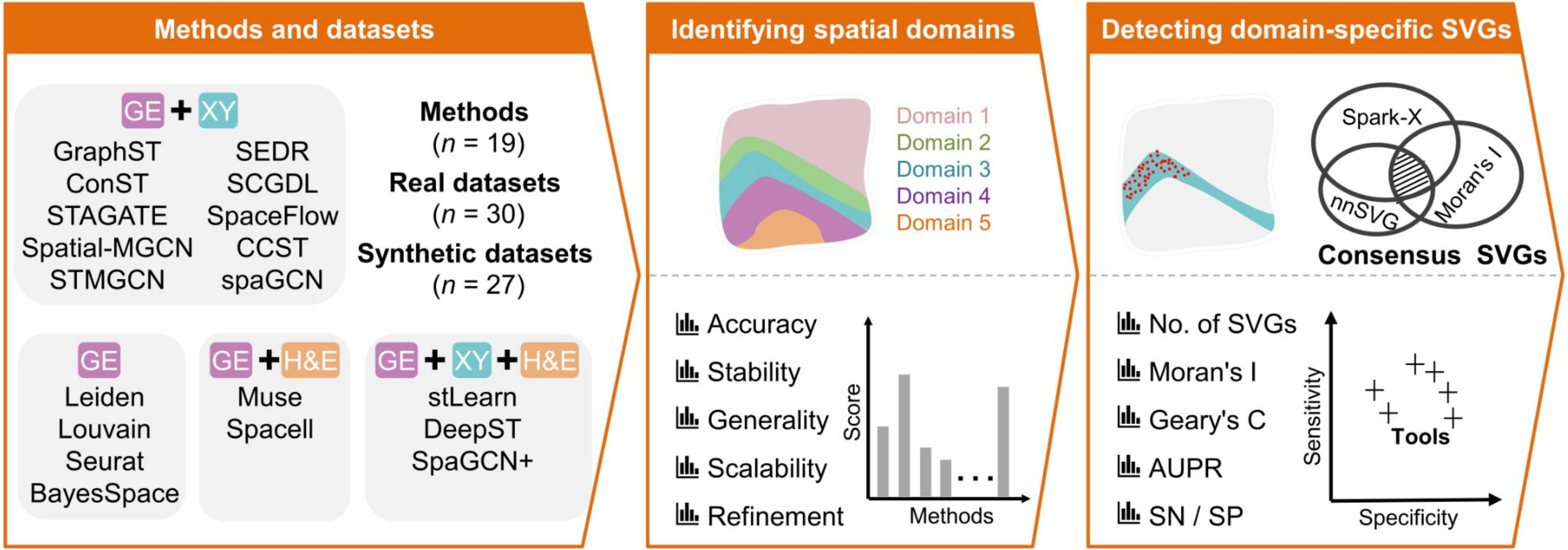
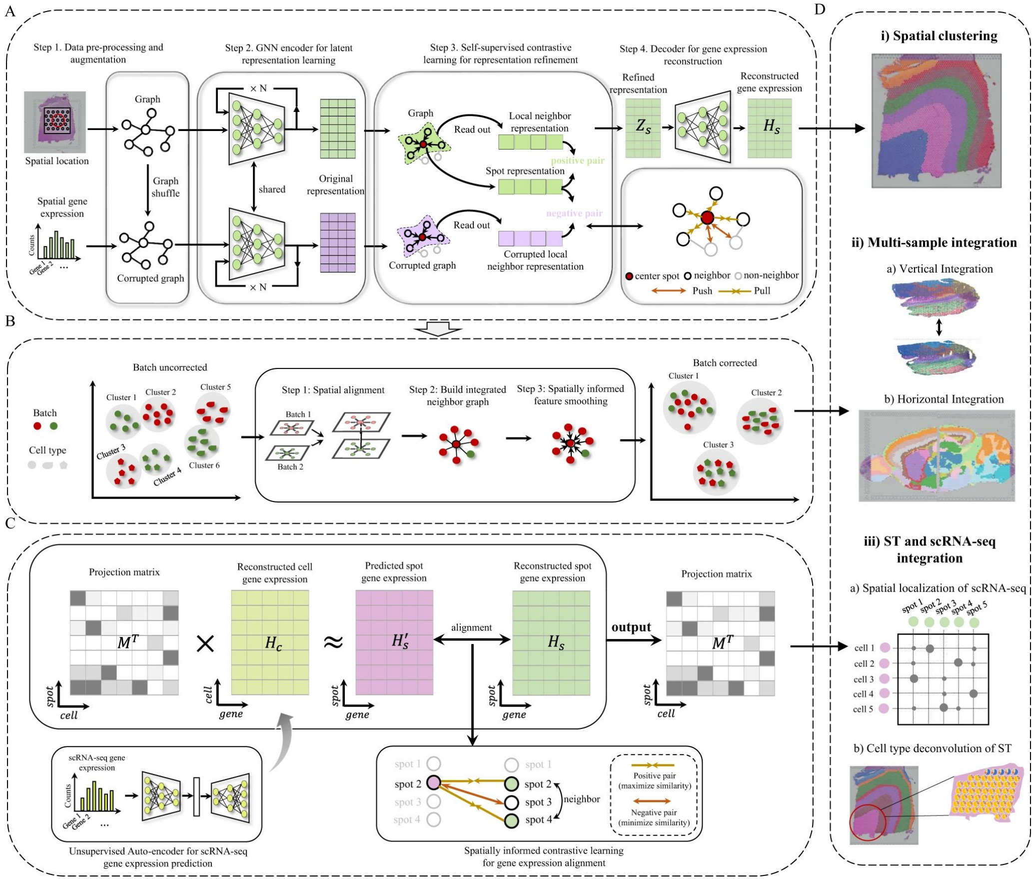
Part I
Methods

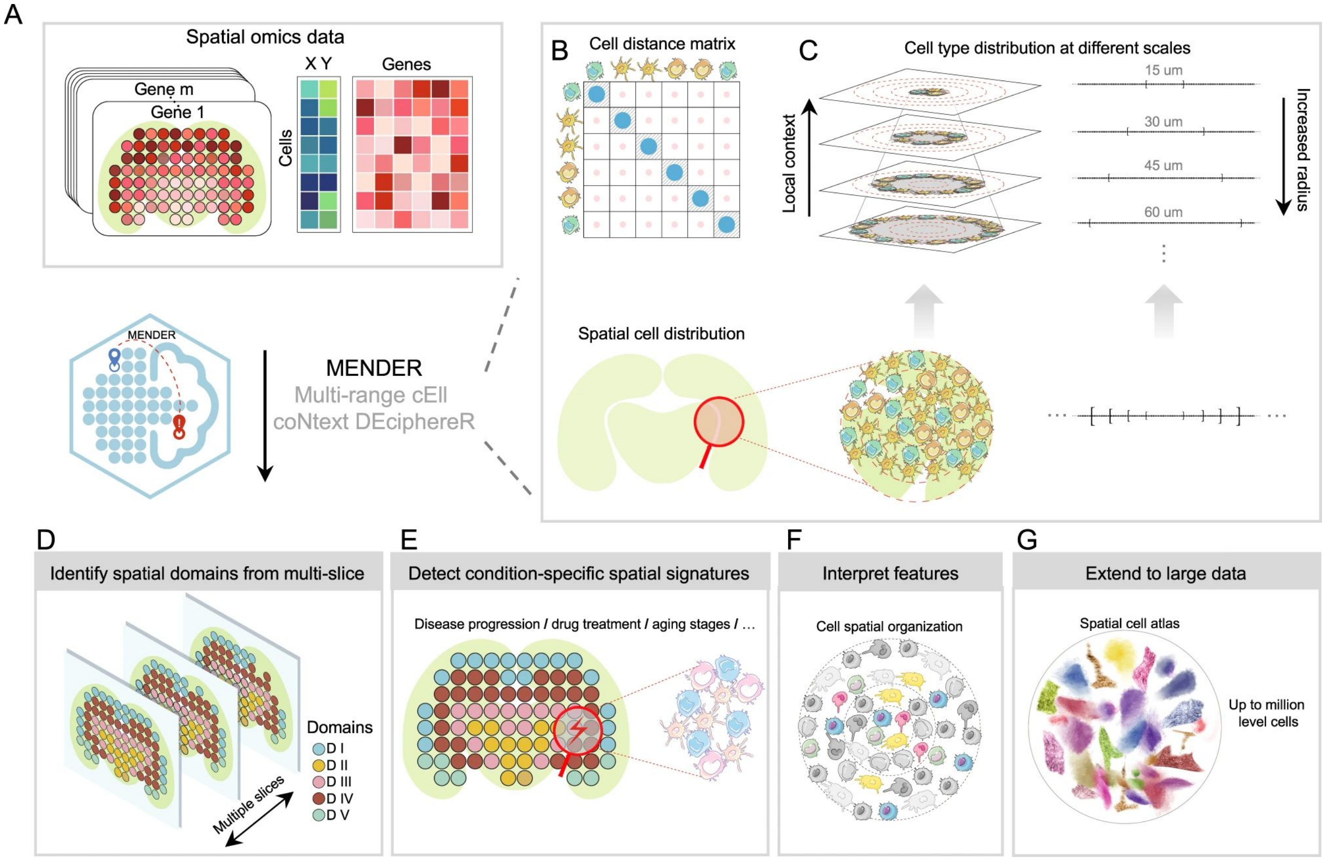


Kang et al., 2023
https://academic.oup.com/nar/article/53/7/gkaf303/8114322



GraphST

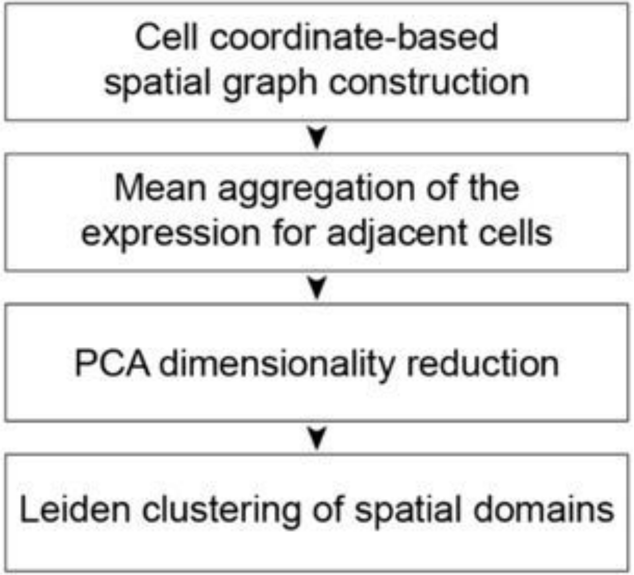
Long et al., 2023
https://academic.oup.com/nar/article/53/7/gkaf303/8114322



MENDER

Yuan, 2024

https://www.nature.com/articles/s41467-023-44367-9

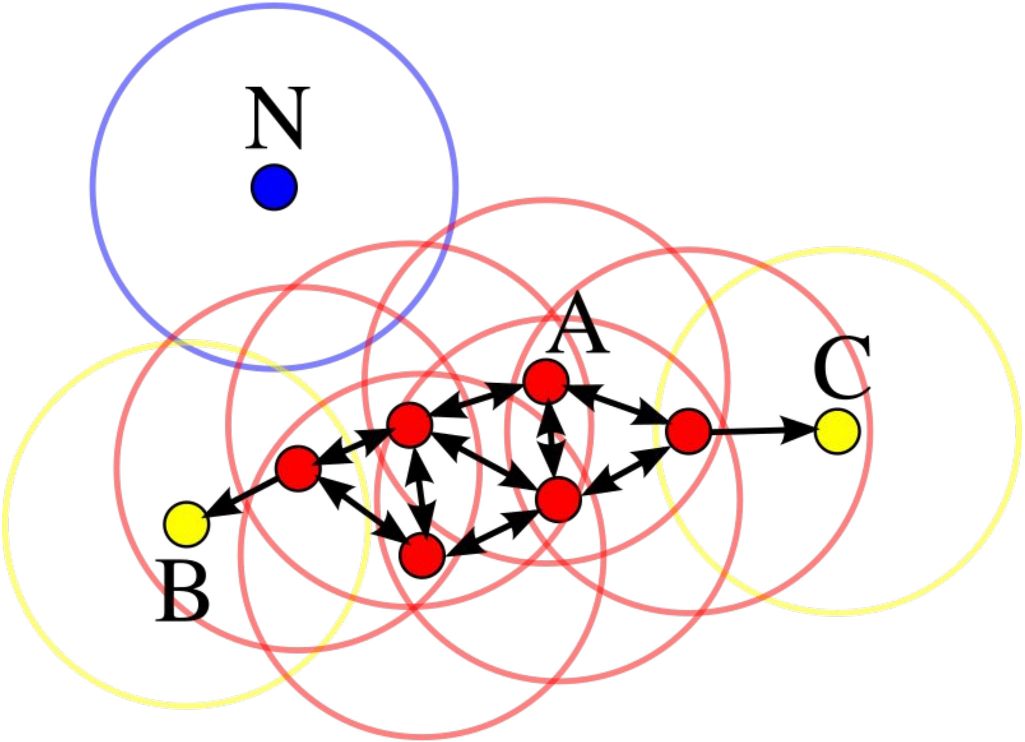


nichePCA

Schaub & Yousefi et al. 2025

https://academic.oup.com/bioinformatics/article/41/1/btaf005/7945104

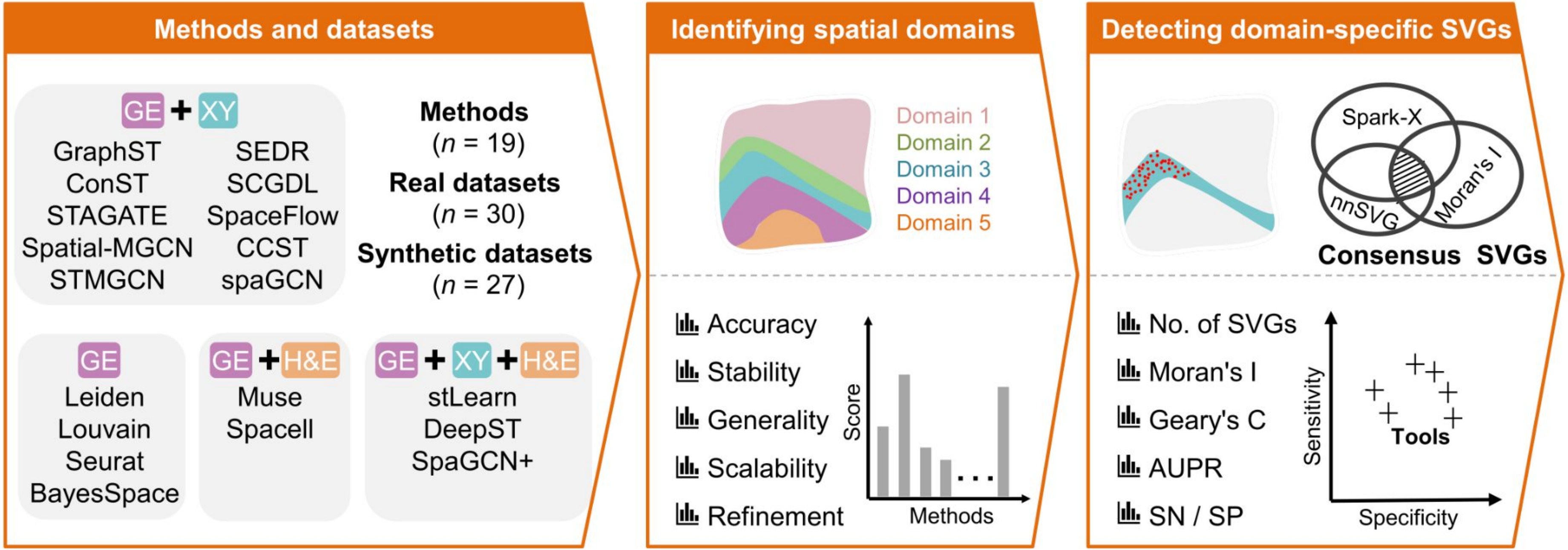
Figure modified from
https://www.biorxiv.org/content/10.1101/2024.12.18.629206v1.full



DBSCAN

Ester et al., 1996

Image source: https://en.wikipedia.org/wiki/DBSCAN



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Software:

- Provide structure for spatial data matrices and accessory data (e.g. UMAPs)
- functions for data visualizations, combining ST data with microscopy data
- standardized workflows for QC (filtering of poorly expressed genes), preprocessing, specialized techniques for ST

Function	R	Python
both well documented and big user communities	Seurat	Scanpy
format	SeuratObject	anndata
specialized workflows for spatial transcriptomics	Giotto	stLearn
Spatially variable gene identification, cell cell interaction	semla	squidpy

Packages

Python

AnnData: https://anndata.readthedocs.io/en/stable/

Scanpy: https://scanpy.readthedocs.io/en/stable/

Squidpy: https://squidpy.readthedocs.io/en/stable/

Spatialdata: https://spatialdata.scverse.org/en/stable/

R

Seurat: https://satijalab.org/seurat/

Giotto: https://rubd.github.io/Giotto_site/

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Questions?